Ajinkya Sanjay Patil

Software Developer

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Professional Summary

Software Developer with a strong background in backend development, specializing in building scalable and high-performance applications. Proficient in Python, Django, PostgreSQL, with experience in designing and optimizing data-driven systems. Skilled in developing microservices and real-time data processing pipelines to ensure efficient and reliable software solutions. Familiar with AWS and cloud-based architectures, with a passion for leveraging modern technologies to drive innovation and efficiency.

Core Skills

Programming Languages: Python, JavaScript
Backend Frameworks: Django,Flask, Node.js
Databases: PostgreSQL, InfluxDB, MongoDB, SQLite, Redis
Server Agent: Telegraf
Messaging & Streaming: Kafka, MQTT
Task Queues & Scheduling: Celery, Celery Beat
IaC & Automation: Docker, Docker Swarm, CI/CD Pipelines
Cloud & DevOps: Basic AWS knowledge (EC2, ECR, RDS, ECS, Code Build, IOT Core, MSK, Elastic Beanstalk, S3)
Others: REST API Development, WebSockets, Microservices Architecture, Version Control (GitHub)

Professional Experience Software

Software Development Engineer 2 May '22 - Present
CarbonMinus (formerly known as Enerlly) Pune

- Deployed Docker Swarm multi-node setups for service orchestration, ensuring system scalability, redundancy and availability.
- Developed and maintained REST APIs using Python and JavaScript, enabling seamless communication between micro services, and integrated WebSockets for real-time alert.
- Configured and managed Apache Kafka clusters for real-time alert and data exchange.
- Optimized system performance by implementing Redis for in-memory data storage and retrieval.
- Used Celery for background data processing in report generation.
- Proficient in Python, JavaScript, Docker, Kafka, PostgreSQL, InfluxDB, MongoDB, Redis, Django, and Node.js for software development.

Software Developer Intern Feb '22 - Apr '22
CarbonMinus (formerly known as Enerlly) Pune

- Learned Django from scratch and applied best coding practices to build scalable applications.
- Refactored existing code to improve efficiency, readability, and maintainability.
- Developed and executed unit tests, ensuring code reliability and created a custom test runner.
- Worked on IoT systems, fetching real-time data from sensors and meters and transmitting it to the cloud.
- Integrated Subscription module with the company's project to automate invoicing, billing and subscription management.
- **Technologies:** Python, Django, PostgreSQL, Celery

Key Projects

P2P Overlay Network for Low-End Devices (POC Phase)

- Designing a peer-to-peer overlay network for low-end and heterogeneous IoT devices to enable remote access, terminal control, OTA updates, and real-time monitoring.
- Gaining hands-on experience with overlay network architecture, NATS for messaging, IP range allocation strategies, and secure device identification.
- Leveraging Ansible for OTA deployments, Django/Python for backend APIs, React.js for frontend, and RabbitMQ/Node.js for device communication workflows.
- Inspired by real infrastructure challenges, the project explores scalable solutions to common issues like connectivity loss, update reliability, and device diagnostics.

Continuous Learning

- Exploring AWS services, including EC2, S3, RDS, ECS, and IoT Core through hands-on practice.
- Improving skills in JavaScript and Node.js with a focus on backend development and microservices.

Education

Bachelor of Engineering Jan '17 - Jan '21
TSSM's Pd. Vasantdada Patil Institute of Technology Pune